

DS4002 Case Study Rubric – Detecting AI-Generated Images

Why am I doing this? The purpose of this case study is for you to apply a complete, end-to-end data science workflow to an image classification problem in a way that is reproducible, well-documented, and easy for others to follow. You will use machine learning methods to analyze and distinguish between different image categories, while demonstrating strong practices in data handling, modeling, evaluation, and communication of results.

What am I going to do? Your task is to build and organize a fully reproducible GitHub repository that walks through the entire data science pipeline from start to finish. This includes properly structuring your data, writing clear and well-organized scripts for each stage of the process, generating and saving relevant outputs, and documenting your work in a clear README file. The repository should be structured so that someone unfamiliar with your project can easily understand, navigate, and reproduce your results step by step.

How will I know I succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Formatting	Goal: Your final GitHub repository must follow this folder structure: -README.md -Data: • Sub-Folder 1: “Orginal_Datasets” • Sub-Folder 2: “Final_Datasets” -Scripts: • All the necessary code files to complete the case study with proper names • Cleaning files, output files, and model files -Outputs: • All the visualizations, tables, or other relevant outputs from the code files with proper names -References: • All the references utilized to help with the context of the case-study and the creation of the scripts files
README.md	Goal: Briefly sumamrize your GitHub repository and project structure using a Markdown file • Include a straightforward overview of

	the project's motivation/context ● Include headers on the structure of the repository ● Include clear instructions on how to go through the repository effectively for those unaware of how to conduct this case-study ● Include all the references used for this project
Data	Goal: Include the datasets used in a CSV format ● Include 2 sub-folders named "Original_Datasets" and "Final_Datasets" ● The "Original_Datasets" sub-folder must have the unprocessed collection of images ● The "Final_Datasets" sub-folder must have the cleaned, processed collection of images
Scripts	Goal: Include all the code used to execute the pipeline - Include every Python code file (either .ipynb or .py) created/used to execute the necessary model and outputs
Outputs	Goal: Show your final results through effective visualizations in a PDF format ● Provide exploratory plots on the data prior to the creation of the model ● Provide visualizations or tables after creating the model to prove the effectiveness of the model. This can include: ○ Confusion Matrices ○ Accuracy Tables
REFERENCES.md	Goal: Include all the references in a Markdown file ● Include any/all references utilized throughout the process of the case-study in a bullet-point format